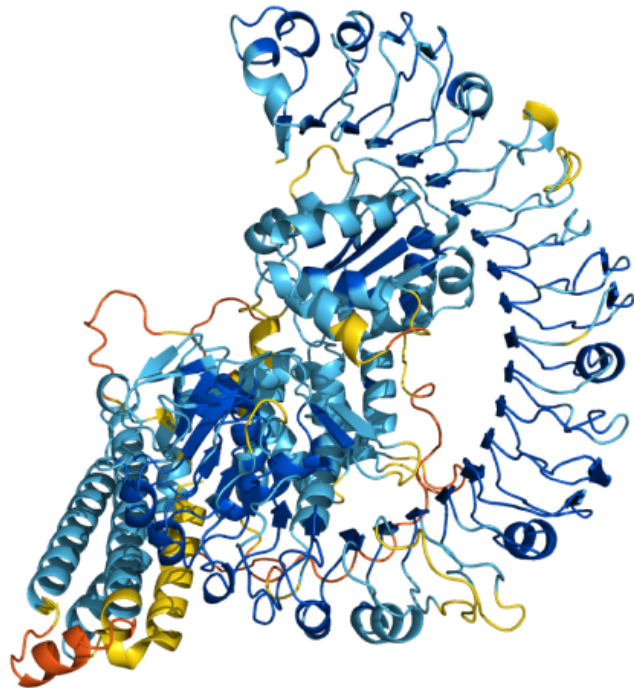


# Protein Function Explorer



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## **Abstract**

Proteins are biomolecules that work together in an organism to support life. Understanding protein structure and how that structure drives protein function will allow us to understand how living organisms work. This has endless applications in many fields, including medicine and agriculture. Understanding protein function requires knowing a protein's structure. How many proteins have a solved structure but an unknown function? Prior to 2021 it was less than 5000. With the development of the AlphaFold artificial intelligence model, this number is already greater than 1.3 million and is predicted to grow to over 100 million by the end of 2022. The next critical step in understanding how genotypes lead to phenotypes is the prediction (and subsequent validation) of protein functions. The goal for this project is to generate a machine learning model that can use inputs from a variety of tools to predict function(s) for proteins without a known function. This will include powerful AI models that do protein function prediction, but do not provide the information in a format that can be utilized by bench scientists, students, educators or health professionals. This gap in the research infrastructure needs to be closed. The proposed Proteopedia Function Explorer will utilize multiple data sources and AI models to generate input variables, in order to produce an overall protein function prediction. The model being developed here is the first step in this process. It will focus on predicting the first number of the Enzyme Commission number, an identifier that describes the function specifically of enzymes (a subset of all proteins).



## **Type of Project**

This project is a multivariable classification problem. We will use a variety of machine learning models and data generated using various tools in order to predict the enzyme commission (EC) number for proteins (specifically enzymes) with unknown functions. Having a predicted EC number is the first step in predicting an enzyme's specific function.

## **Data**

### **Data Source**

The current data set was collected from a variety of sites (DATA 445, Fall 2021). A key source of data was Protein Databank (PDB, <https://www.rcsb.org/>), a public repository of proteins with solved structures (RCSB Protein Data Bank, 2021). Since the goal of this project is to build a machine learning model that can predict protein function from only a protein sequence and/or structure, it is important to have a starting data set that is well validated. The PDB is a single source for proteins contains peer-reviewed solved structures and also has annotation about their function. Pfam categorical descriptors are also included in the data set.

Many existing Deep Learning and Artificial Intelligence models produce protein function predictions (and these predictions utilize skills well beyond the level of our team). Upon examining these models in detail, it became clear that most produce function predictions by outputting a list of Gene Ontology (GO) terms. GO terms were developed as machine learning became more prominent in science applications. Every protein has a collection of associated GO terms, and each describes an aspect of a protein's molecular function, biological function or cellular location. The collection of GO terms gives a snapshot of a protein's function at many levels, including information that should be useful for predicting EC number. GO terms were created to be machine-learning compatible, and consist of two parts: a human-readable phrase (e.g. transferase activity) and a GO number (e.g. GO:0016740). Since the purpose of this project is to use relevant data to predict protein function, we will be focusing heavily on GO terms.

Any manually curated gene ontology (GO) molecular function terms available on the public UniProt site (<https://www.uniprot.org/>) were added to the dataset previously. Binary variables were engineered from these GO terms, for the seven major classes of enzymes. The first EC number (here referred to as ec1) has possible designations: ec1 = oxidoreductase, ec2 = transferase, ec3 = hydrolase, ec4 = lyase, ec5 = isomerase, ec6 = ligase, and ec7 = translocase. Binary GO variables were created, by searching the full list of GO terms for each protein. If the term was present the binary variable was scored as "1", otherwise it was scored as "0". Additional GO term predictions will be added to the dataset this semester.

### **Data Description**

The existing data in this project was generated from a variety of information compiled from various websites. These are provided as highly nested dictionary files, xml files, strings, objects, series and other formats. A key challenge in this project is the large number of high

dimensionality categorical variables. Feature engineering will be key in order to work effectively with this data set. The data has already been split into train (80%) and test (20%) subsets, with only train data being used for modeling and hyper-parameter tuning.

### Variable Descriptions

| <b>Table 1. Variable names and descriptions</b> |                      |                                                                                                                                                                                                                 |                  |
|-------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Columns Index</b>                            | <b>Variable Name</b> | <b>Description</b>                                                                                                                                                                                              | <b>Data Type</b> |
| 0                                               | PDB_ID               | Unique Identifier for solved protein structure                                                                                                                                                                  | String           |
| 1                                               | PDBchain             | Unique identifier for each protein chain in a PDB file (letters only, starting with A)                                                                                                                          | String           |
| 2                                               | len                  | The number of amino acids in the protein chain described                                                                                                                                                        | Integer          |
| 3                                               | method               | Method used to solve the crystal structure                                                                                                                                                                      | String           |
| 4                                               | resol                | The resolution of the structure, in Angstroms                                                                                                                                                                   | Float            |
| 5                                               | rfac                 | Calculated value that indicates the quality of the information in a crystallographic study. May be blank if the method used is not x-ray crystallography                                                        | Float            |
| 6                                               | freerfac             | Calculated value that indicates the quality of the information in a crystallographic study. Based on a subset of the data set aside for the calculation of free R, and not used in the refinement of the model. | Float            |
| 7                                               | ec                   | The Enzyme Commission (EC) number of the individual protein                                                                                                                                                     | String           |
| 8                                               | aaseq                | The amino acid sequence of the protein, in one-letter amino acid code                                                                                                                                           | String           |
| 9                                               | accession            | The accession number, mainly for the UniProt database                                                                                                                                                           | String           |
| 10                                              | PDBID                | Duplication of the PDB number above—generated when pulling the Pfam data, and used to ensure Pfam information appends to the correct rows of the dataset.                                                       | String           |

|                         |                                    |                                                                                                                                                                                                                                          |         |
|-------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 11-37                   | Pfam1 - Pfam26                     | A set of columns used to store the Pfam family identifications reported back from the Pfam database. Forty columns were used as the total number varies and cannot be known in advance. Forty columns allowed code to run without error. | String  |
| 38-41                   | ec1 - ec4                          | Enzyme Classification numbers (from the format 1.4.3.2)                                                                                                                                                                                  | Integer |
| 42                      | Entry                              | UniProt identifier for additional data collection                                                                                                                                                                                        | String  |
| 43                      | Entry name                         | UniProt name                                                                                                                                                                                                                             | String  |
| 44                      | Gene names                         | Abbreviated gene names (useful for additional data collection)                                                                                                                                                                           | String  |
| 45                      | Gene ontology (molecular function) | GO identifiers associated with molecular function of the protein                                                                                                                                                                         | String  |
| 46                      | EC number                          | EC number again (to verify dataset integrity as do various merges and concatenations)                                                                                                                                                    | String  |
| 47                      | Gene ontology IDs                  | The full list of GO identifiers associated with the protein (molecular function, cellular location, biological function)                                                                                                                 | Integer |
| 48                      | Sequence                           | The amino acid sequence of the protein                                                                                                                                                                                                   | String  |
| 49 - 83                 | GO1 - GO35                         | A split of the list of GO names and numbers, separated to have only one pair per protein                                                                                                                                                 | String  |
| 84 - 153 (Even Columns) | GO1Str - GO35Str                   | The human readable GO name (string) from GO1-GO35                                                                                                                                                                                        | String  |
| 84 - 153 (Odd Columns)  | GO1Int - GO35Int                   | The machine-readable GO number (integer) from GO1-GO35                                                                                                                                                                                   | Integer |
| 154                     | Activity Count                     | Count of molecular activities per protein                                                                                                                                                                                                | Integer |
| 155 - 167               | Activity1 - Activity13             | GO molecular activity numbers for each protein (max 13)                                                                                                                                                                                  | Integer |
| 168                     | Oxidoreductase                     | Binary classifier searching for GO terms containing the EC function listed                                                                                                                                                               | Integer |

|     |             |                                                                            |         |
|-----|-------------|----------------------------------------------------------------------------|---------|
| 169 | Transferase | Binary classifier searching for GO terms containing the EC function listed | Integer |
| 170 | Hydrolase   | Binary classifier searching for GO terms containing the EC function listed | Integer |
| 171 | Lyase       | Binary classifier searching for GO terms containing the EC function listed | Integer |
| 172 | Isomerase   | Binary classifier searching for GO terms containing the EC function listed | Integer |
| 173 | Ligase      | Binary classifier searching for GO terms containing the EC function listed | Integer |
| 174 | Translocase | Binary classifier searching for GO terms containing the EC function listed | Integer |

### Data visualizations

The goal for this project is to predict enzyme commission (EC) numbers for proteins of unknown function. Not all classes of enzymes are found in equal numbers in nature, and in addition a new class of enzymes (EC 7) was added to the EC system in 2018. This new class has fewer members, because there was not a recognized need for it until 2018. The EC number distribution is definitely not balanced, with EC 2 and EC 3 having significantly more members than the other classes (Fig. 1). The newest class, EC 7, has only had proteins being classified into it since 2018 and so there are very few entries.

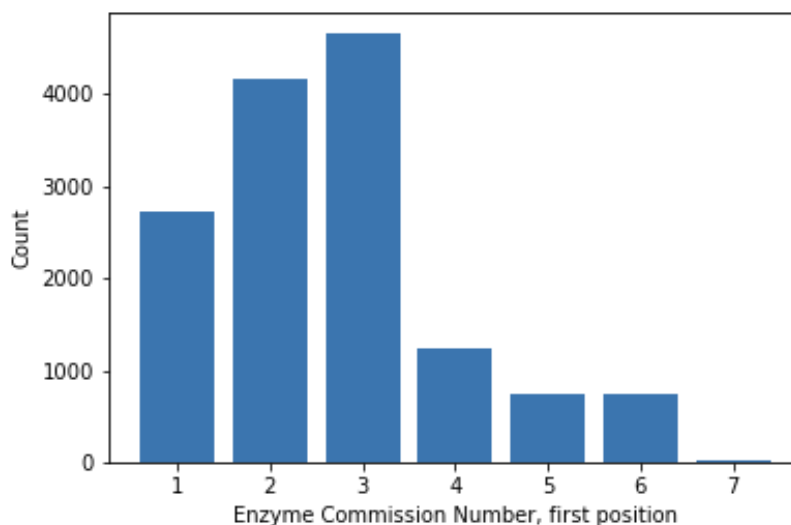


Figure 1 Enzyme Commission number distribution in the dataset

The Pfam families represent 26 columns in the final modeling dataset. An examination of the data distribution reveals that most proteins have 5 or fewer families associated with the entry (Figure 6). We will be unable to use Pfam inputs with most of the models as they are high dimensionality categorical variables. Possibly feature engineering could allow these variables to be used as model inputs more effectively.

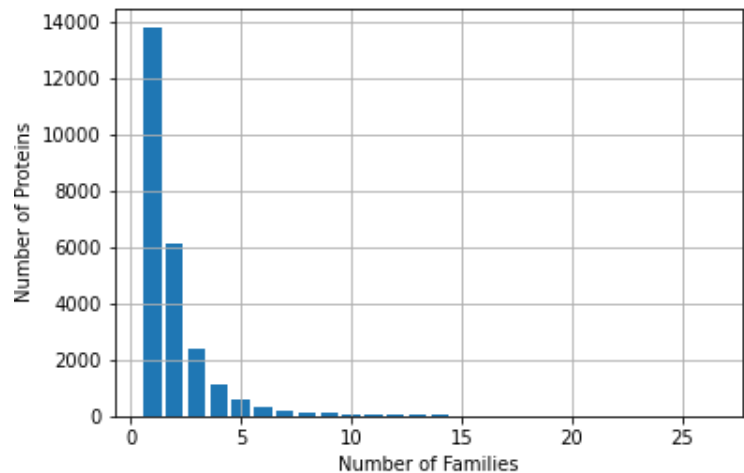


Figure 2. Distribution of number of Pfam families associated with each individual protein in the dataset

The molecular GO activities associated with each individual protein varied from 1 to 13 (Figure 7). Most proteins had at least one GO activity associated with it, with up to 5 being common. It was rare for a protein to have 6-13 GO activities associated with it.

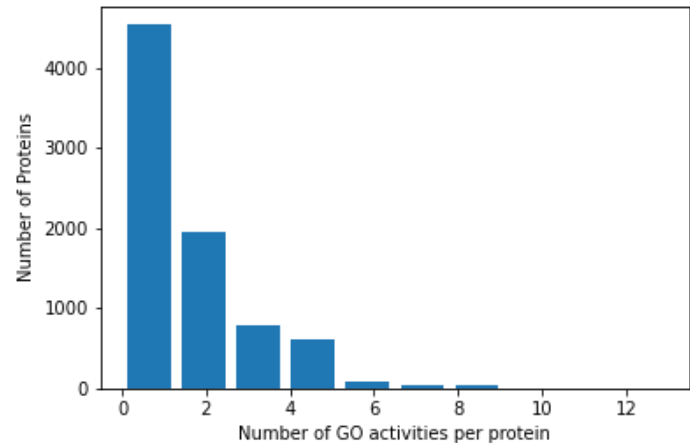


Figure 3. Distribution of number of GO activities associated with each individual protein in the dataset

## Data Sources

1. PISCES: Creates culled lists of proteins, allowing users to select varying levels of sequence identity between proteins. Sequence identities for PDB sequences are determined by creating a hidden Markov model for every unique PDB sequence with the program HHblits and searching the resulting collection of HMMs with each individual HMM with the program HHsearch (Wang, 2003). PISCES can therefore provide meaningful results at low sequence identity (15-30%). Publicly available online at <https://dunbrack.fccc.edu/piscs/>.
2. The Protein Databank (PDB): Through an internet information portal and downloadable data archive, the PDB provides access to 3D structure data for large biological molecules (proteins, DNA, and RNA) (RCSB Protein Data Bank, 2021). These are the molecules of life, found in all organisms on the planet. RCSB PDB (Research Collaboratory for Structural Bioinformatics PDB) operates the US data center for the global PDB archive, and makes PDB data available at no charge to all data consumers without limitations on usage. Publicly available online at <https://www.rcsb.org/>.

3. Pfam: The Pfam database (Mistry, 2020) is a large collection of protein families, each represented by multiple sequence alignments and hidden Markov models (HMMs). Proteins are generally composed of one or more functional regions, commonly termed domains. Different combinations of domains give rise to the diverse range of proteins found in nature. The identification of domains that occur within proteins can therefore provide insights into their function. Pfam also generates higher-level groupings of related entries, known as clans. A clan is a collection of Pfam entries which are related by similarity of sequence, structure or profile-HMM. The data presented for each entry is based on the UniProt Reference Proteomes. Publicly available online at <http://pfam.xfam.org/>
4. Online machine learning tools will be utilized to generate additional .json outputs of Gene Ontology (GO) terms. This will include NetGO (<https://issubmission.sjtu.edu.cn/netgo/>) , and possibly also DeepFRI (<https://beta.deepfri.flatironinstitute.org/>) if an API interface can be established..

## **Importance of this Project**

The goal of this project is to continue refining our machine learning models built previously, to provide accurate prediction of protein function. Proteins are essential for life, and understanding how they work is critical in modern medicine, scientific industry and agriculture. The data revolution has led to an explosion in scientific data. This began with sequencing the genomes of thousands of organisms, which provides us with almost unlimited numbers of gene sequences. This sequence information can easily be used to output the sequence of all the proteins these genes encode. The Protein Structure Initiative set out to solve structures for as many proteins as possible, resulting in deposition of thousands of protein structures in the PDB. The goal was to get enough information about protein structure to begin predicting protein function.

The development of the AlphaFold AI system this past year by Deep Mind and Google suddenly means that using only computation, protein structures can be predicted with accuracies as good as the information from hands-on experimentation (Jumper, 2021). In July 2021, the AlphaFold database contained approximately 350,000 solved structures, currently contains 1.3 million solved structures and is predicted to grow to over 100 million structures by the end of 2022. The next step is to use these structures to predict protein function. Multiple AI tools have been developed and can provide powerful predictions from an AlphaFold structure. Unfortunately, the outputs from these systems are not in formats understandable for the majority of scientists and physicians with the training to validate these predictions and use them to solve real world problems. This is referred to by the National Science Foundation (NSF) as a mid-scale research infrastructure gap (a gap between large and complex systems and individual scientists performing specific experiments). Developing mid-scale research infrastructure is one of NSF's Ten Big Ideas that hopes to prevent missed opportunities from leaving essential science undone (National Science Foundation, n.d.).



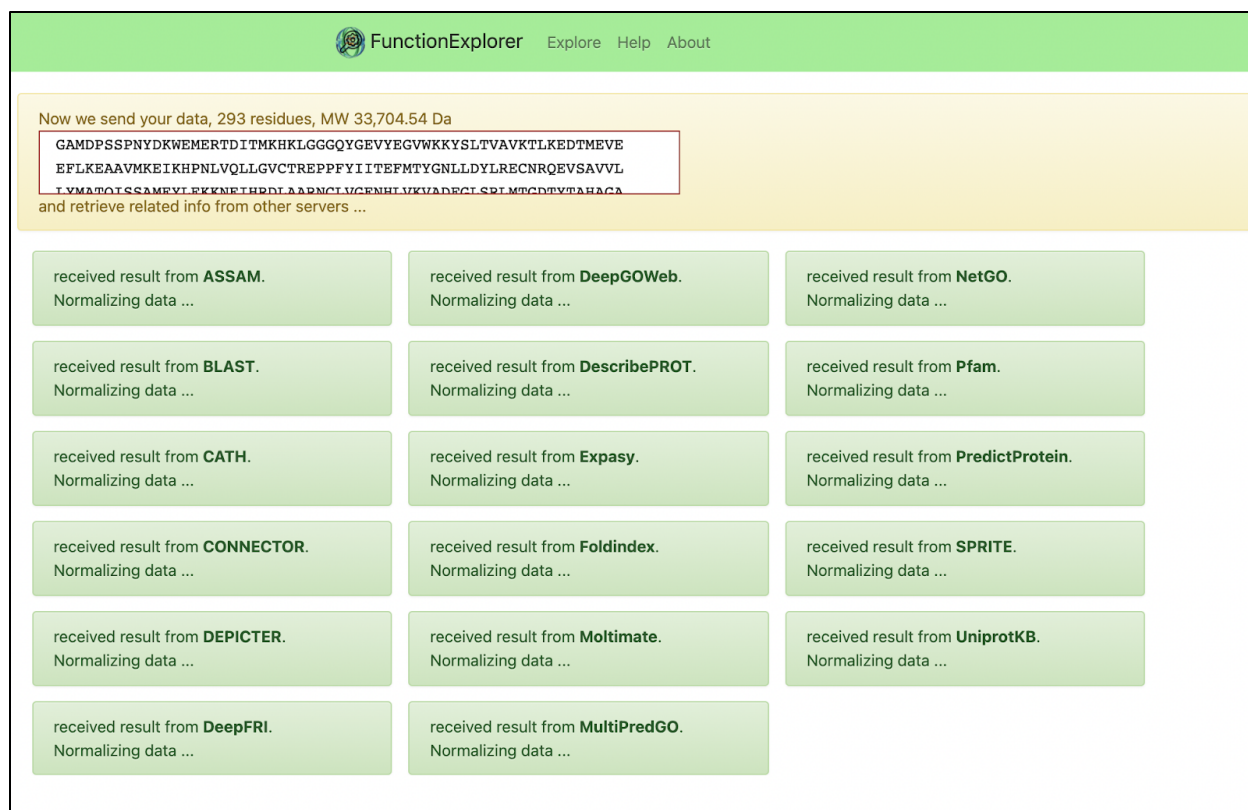


Figure 4 Screenshot of the Proteopedia Function Explorer prototype website

The goal for the machine learning model(s) being developed here is to work to close this mid-scale infrastructure gap. Ultimately, we will generate the Proteopedia Function Explorer, a public facing website (Prilusky, 2021) that will allow users to input the sequence of a protein (Figure 4).

AlphaFold can then be queried and used to determine the protein’s structure (Jumper, 2021). The sequence and structure information together can be used as inputs for a variety of powerful and publicly accessible tools (see data sources above for details) to generate a set of data about the protein’s possible function. The model being developed here will then be used to clean, assemble and integrate the results following (to be) established rules (Figure 5). There may ultimately also be some user tuning allowed. An understandable prediction will be provided to the user for the most likely function of the protein, with a long-term goal of also providing information that will allow users to perform wet lab experiments to try and validate and/or apply the prediction.

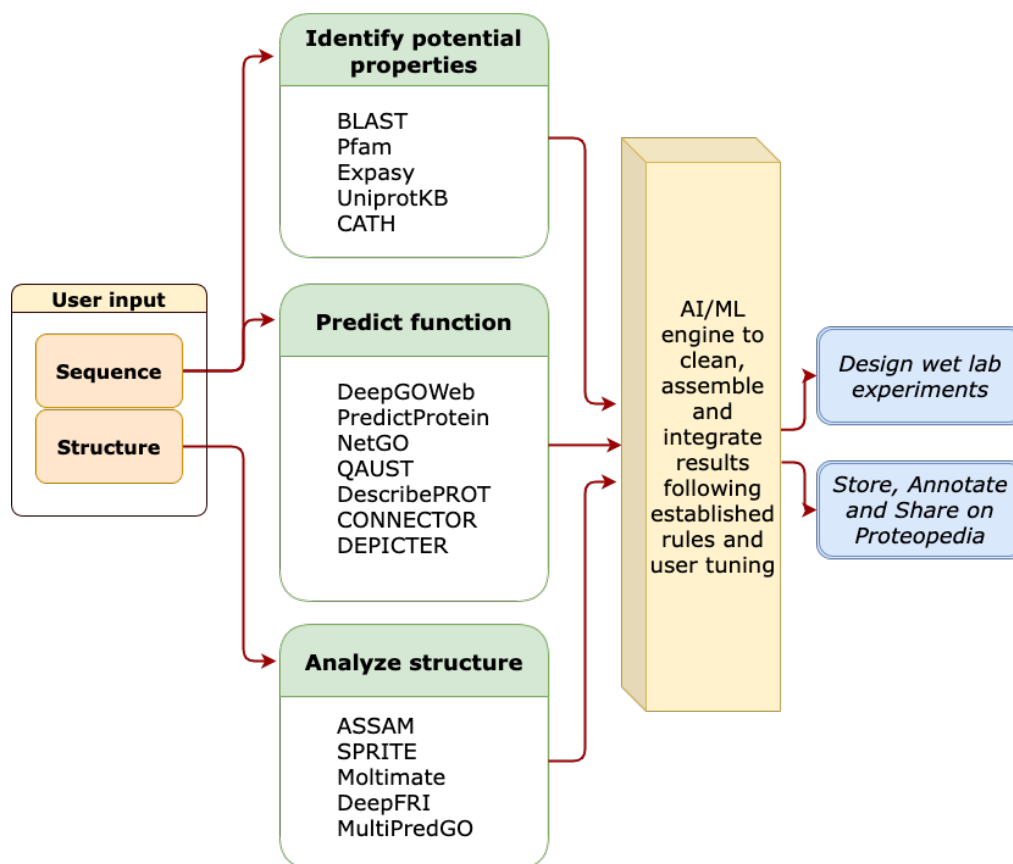


Figure 5. Workflow for generating predictions for users of the Proteopedia Function Explorer

## Research Plan

Phase 1: Rebuilding the existing models, as we know much more now about building predictive models. We know there are flaws in our models and want to revisit them and improve the logic and data handling for those models.

Phase 2: Expanding the data set. Outputs from the NetGO server (<https://issubmission.sjtu.edu.cn/netgo/>), which represents proteins using Evolutionary Scale Modeling (ESM) embedding and then uses Logistic Regression (LR) to train a new model, LR-ESM. If an API or batch output can be accessed then the DeepFRI server (<https://beta.deepfri.flatironinstitute.org/>) will also be used to generate GO term predictions. DeepFRI uses graph convolutional networks with language model features.

Phase 3: Variable engineering. Many of the variables are high dimensionality categorical variables. We will need to engineer features from these variables in order to have usable input variables. There will also need to be dimensionality reduction in this engineering process.

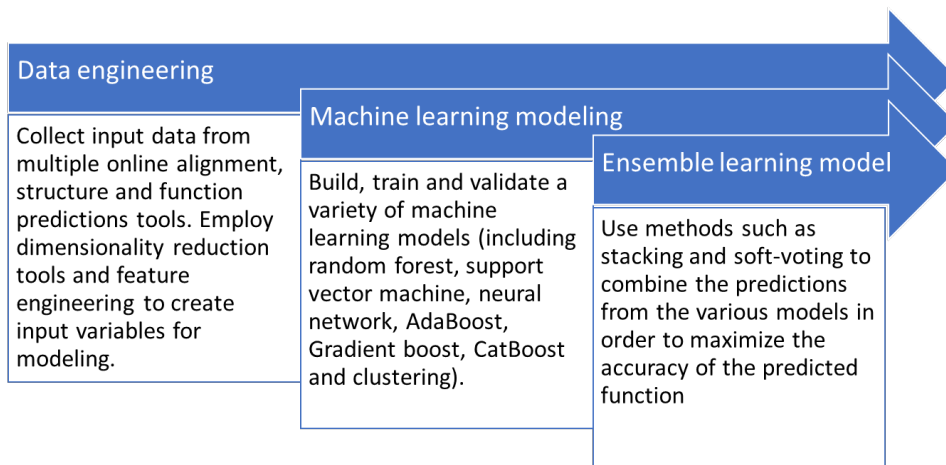


Figure 6. Workflow for developing the Proteopedia Function Explorer machine learning models to use in retrieving, collating and predicting the most likely protein function(s)

Phase 4: Model development. A variety of new models will be built, including random forest, neural network, SVM, AdaBoost and Gradient Boosting. The goal is to try multiple model types, with more than one likely being used for the final predictions. The models will undergo variable selection and hyper-parameter tuning.

Phase 5: Ensemble modeling. We will ultimately likely get the best predictions using an ensemble model, possibly with a soft-voting strategy (this will be finalized based on predictions). The best models from phase 4 will be used to create an ensemble learning model.

Phase 6 and beyond: Additional data generation and model refinement. This work will be beyond the scope of this course-based project. But project will continue after DATA 448 ends, hopefully with funding from the Binational Science Foundation (grant application pending).

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